Large Models for Resource Allocation in Edge Computing Power Networks

Liyan Sui, Ke Zhang, Fan Wu, and Xiaoyan Huang

Abstract

Computing Power Network (CPN), as an integration of heterogeneous computing resources, has turned to be difficult to cope with the rapid growth of end users and strict task delay constraints due to the possible long-distance transmission between its service facilities and remote users. To overcome this problem, we propose an Edge Computing Power Network (ECPN), which reduces long-distance transmission overhead, optimizes resource utilization, and improves system performance by fully utilizing edge computing resources. In the area of resource scheduling, traditional machine learning is widely used due to its fast problem solving capabilities. However, traditional machine learning faces challenges in ECPN, particularly with handling heterogeneous tasks and adapting to highly dynamic network conditions. To address these challenges, we propose a large model-enabled ECPN framework that enhances the capabilities of ECPN in task offloading and resource allocation while optimizing the deployment of large models in the framework. To demonstrate the advantages of this framework, we introduce a smart transportation application scenario and employ a large language model based graph multi-agent deep reinforcement learning algorithm to determine optimal task matching strategies that balance energy consumption and privacy protection in the ECPN. Experimental results demonstrate that the algorithm applied in this framework reduces energy consumption, decreases the convergence time for task matching, and enhances privacy protection in ECPN.

Introduction

In the 5G/B5G era, Computing Power Network (CPN) is a service paradigm that connects computing resource devices from both core networks and edge sides to achieve ubiquitous and powerful processing capabilities [1]. However, when a CPN component relies on long-distance or low-capacity network connections, the task data offloading latency may increase, potentially hindering the efficiency and responsiveness of the service process [2].

To reduce transmission latency while ensuring task processing efficiency, we propose an Edge Computing Power Network (ECPN), which improves resource utilization and task processing efficiency by integrating the computing resources of edge nodes close to the task generator. Although traditional Machine Learning (ML) has been widely used in resource scheduling in various networks, it still faces severe challenges being applied in ECPN with large-scale edge nodes and dynamic network states. These challenges include inefficiencies in processing concurrent heterogeneous tasks and the lack of adaptive resource allocation and energy-efficient scheduling strategies tailored to diverse task requirements.

Considering these challenges, we introduce large models into ECPN to leverage their advanced capabilities in global optimization and adaptive control to further improve task scheduling efficiency and resource utilization. Large models have a complete structure and a large number of parameters, and by training on large-scale datasets, they can automatically extract features and identify patterns without relying on predefined features [3]. This capability makes large models more adaptable than traditional ML models when handling heterogeneous tasks and processing large-scale data in complex and evolving scenarios [4]. The global optimization and adaptive adjustment capabilities of large models improve task processing efficiency and reduce system response time, especially in parallel multi-task scheduling and timely adaptation to dynamic environments [5]. For example, Rong and Rutagemwa [6] applied large models to the intelligent control of a 6 G global integrated network to optimize internet of things services through efficient resource management and adaptive strategies. Zhang et al. [7] pioneered a diversity-driven proactive caching scheme to optimize network load and resource efficiency, which provided inspiring insights for large-scale model deployment in edge environments. However, while these works have made progress in network optimization, they have not explore the integration of large models into ECPN with large-scale edge nodes and highly dynamic environments.

This paper introduces large models in ECPN for the first time to improve the efficiency and accuracy of resource allocation for edge computing nodes. Large models enhance the ability of ECPN to extract critical information from heterogeneous data through global feature extraction. Simultaneously, an adaptive adjustment mechanism optimally reallocates resources in real time, enabling effective response to dynamic changes in edge computing resources and task requirements [8]. Furthermore, the ECPN leverages the parallel processing capabilities of large models

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to concurrently manage multiple heterogeneous tasks, thereby improving performance and optimizing resource utilization in dynamic environments [9].

Although large models demonstrate significant potential across various domains, their deployment on edge devices within the ECPN framework faces challenges such as resource constraints, real-time responsiveness, and the maintenance and updating of models [10]. To address these issues, we leverage advanced techniques into the large model-enabled ECPN to enhance real-time processing, improve model reuse, and ensure efficient maintenance in resource-constrained settings. First, model compression technology is used to effectively reduce the complexity of calculations and storage requirements, making large models suitable for resource-constrained edge devices. Second, distributed inference technology distributes inference tasks to multiple devices for collaborative processing to meet the needs of real-time task response. In addition, incremental learning technology allows large models to continue learning new data on top of the original foundation, adapting to changes without retraining, thereby improving model reusability. To demonstrate the effectiveness of the applied techniques and highlight the advantages of the large model-enabled ECPN, we design a multi-agent deep reinforcement learning (MADRL) algorithm based on large language models (LLMs) to optimize the privacy-preserving task matching strategy for ECPN in smart transportation systems.

The rest of this paper is organized as follows. The section "Large Models Enabled Edge Computing Power Networks" presents the architecture and key aspects of large model integration in the ECPN. We formulate a LLM-enabled task matching problem to trade the energy and privacy in the section "LLM-Enabled Privacy-Preserving Task Matching in ECPN." Then we evaluate the performance of our proposed architecture using numerical results in the section "Numerical Results." Finally, the section "Conclusion" concludes this paper.

> Large models enhance the ability of ECPN to extract critical information from heterogeneous data through global feature extraction.

Large Models Enabled Edge Computing Power Networks

In this section, we propose the large models-enabled ECPN framework, and demonstrate applications, benefits, challenges, and corresponding solutions of the framework.

The Architecture of Large Model-Enabled ECPN

Fig. 1 shows the architecture of the proposed large models empowered ECPN. While CPN integrates computing resources from both core networks and edge locations to enhance task processing capabilities, ECPN specifically concentrates on optimizing the use of edge nodes to facilitate the deep integration of communication and computing capabilities at the edge. ECPN improves the responsiveness of the system to tasks in different environments through a flexible edge resource awareness. The architecture of ECPN is composed of an edge computing power plane and an application plane. In the edge computing plane, the

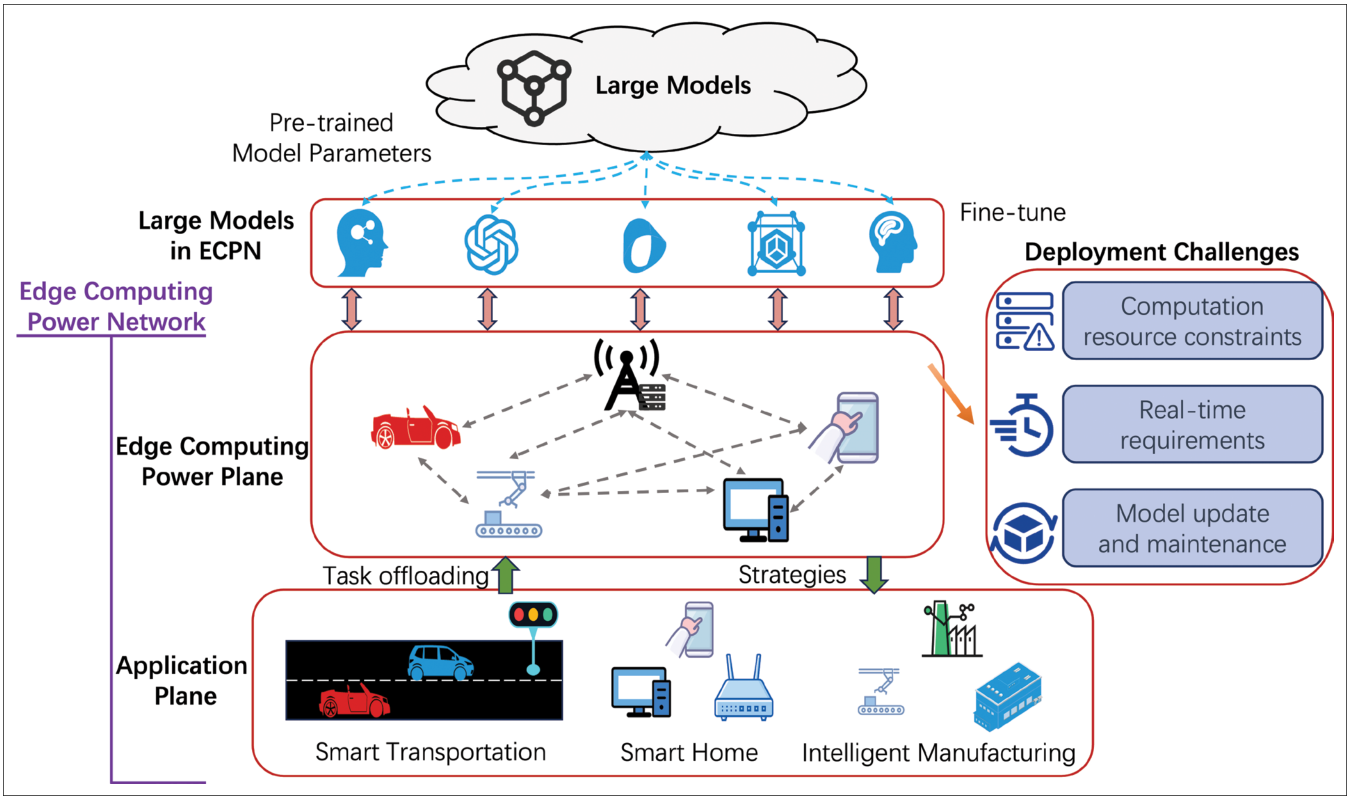

FIGURE 1. The architecture for large model empowered ECPN.

In ECPN, although traditional ML algorithms can process some tasks in parallel, they are usually limited to local tasks, making it difficult to achieve efficient parallel processing of multiple heterogeneous tasks.

computing power of connected edge nodes forms a pooled resource to provide computing services to end devices. The edge computing nodes include base stations (BSs), MEC servers and the end devices. The application plane consists of tasks and requirements for different application scenarios, such as smart transportation, smart homes, and smart manufacturing, offloading these tasks to the edge computing power plane for processing. Edge devices perform dual roles in the network. When tasks are generated, these devices request computing resources from other nodes to meet processing demands. Conversely, when idle and equipped with available resources, they provide their computing power to support tasks generated by other devices.

Based on the ECPN architecture, we further explore how to effectively integrate large models to optimize task scheduling and resource allocation. In ECPN, large models are deployed by cloud-edge collaboration, where the cloud is responsible for training large models and distributing the pre-trained model parameters to each edge node. The edge nodes locally fine-tune the large model according to the task requirements and available computing resources to optimize the inference capability [11]. We use model compression and distributed inference techniques on edge device to improve resource utilization and save computing resource on individual edge nodes. With this approach, large models in ECPN are reused and maintained through incremental learning, enabling them to adapt to task requirements and data changes in real world scenarios. Incremental learning allows the model to learn new data from evolving environments continuously without retraining entirely. This approach reduces the costs associated with model updates and ensures that the model can adapt to dynamic task requirements and variations in environmental data across diverse scenarios.

Application for Large Model-Enabled ECPN

Large models enable ECPN to improve the decision-making ability and performance of the system through more accurate data analysis, and achieve improvements in the efficiency and responsiveness of resource allocation. The main applications of the large models in ECPN are as follows.

Smart Home: Smart homes connect various devices, such as sensors, lighting systems and cameras, through smart homes, automating services and offering personalized experiences to enhance user comfort. In this connected environment, the ECPN is able to leverage idle computing resources from devices like smartphones and computers to address additional needs for processing environmental data and executing collaborative tasks. However, when combined with traditional ML methods relying on predefined features and local information, ECPN faces challenges in meeting diverse and dynamic requirements, including handling heterogeneous device interactions, adapting to varying user preferences, and processing large volumes of sensor data. In comparison, large models have powerful data learning capabilities and can extract global features from heterogeneous data generated by multiple devices and sensors in a smart home. Leveraging the processing power of large models, ECPN effectively utilizes distributed computing resources to analyze device interactions, user behavior, and environmental data. This enables optimized resource scheduling, accurate user need predictions, and real-time device adjustments, enhancing smart home performance and user experience.

Smart Transportation: Smart transportation is a system with large-scale management, covering multiple management areas and interrelated block road networks, and requiring the processing of large environmental data. In smart transportation systems, ECPN leverages computing resources from devices like vehicles and roadside units to manage tasks such as traffic control, route optimization, and accident detection. With the resource integration capabilities of ECPN, the system adapts to traffic fluctuations and the flexible distribution of computational tasks to moving vehicles and connected devices, enhancing the immediacy of task processing and the efficiency of resource allocation. When ECPN is combined with traditional ML methods, which focus on specific regions with constrained data related to those regions, it struggles to achieve global optimization across multiple areas. Contrastingly, large models have powerful global feature extraction and adaptive capabilities, and can quickly capture changes in a wide range of heterogeneous data and optimize scheduling strategies in real time. Therefore, ECPN combines with the advantages of large models to achieve dynamic task allocation and resource allocation adjustments, enabling the system to operate efficiently in multiple regions, complex road networks, and rapidly changing traffic flows, improving the scheduling efficiency and adaptability of smart transportation systems.

Intelligent Manufacturing: Intelligent manufacturing systems use intelligent algorithm to achieve differential production management, covering multiple production departments and various types of equipment, and emphasizing rapid response capabilities. In this scenario, ECPN can provide computing services for applications such as real-time device monitoring and production process optimization by scheduling edge computing resources, supporting the needs of smart manufacturing. As these tasks have different computing requirements and response times, they place high demands on resource allocation and task scheduling in the system. In ECPN, although traditional ML algorithms can process some tasks in parallel, they are usually limited to local tasks, making it difficult to achieve efficient parallel processing of multiple heterogeneous tasks. Compared with this, the large model has powerful parallel processing capabilities, and can schedule multiple heterogeneous tasks from different production equipment, sensors and control systems, and dynamically allocate computing resources based on task priorities and real-time requirements. Combined with a large model, ECPN can efficiently respond to the diverse needs of intelligent manufacturing in terms of different production departments and a wide range of equipment, reducing task execution time and improving overall system performance to meet the requirements of rapid response.

Benefits for Large Models in ECPN

Benefiting from the powerful feature extraction capabilities and adaptability of large models in ECPN, the network can achieve the following benefits.

Improving Resource Utilization: As the number and diversity of edge devices in ECPN expand, the requirements for resource management and scheduling escalate to preserve network efficiency and ensure responsiveness. Therefore, improving the resource utilization of ECPN has become critical to ensure that the growing demand for tasks can be addressed and resource waste minimized, improving the performance of the network. Compared with the traditional ML, the large model has the ability to perceive the state of computing nodes and task requirements in all aspects, understand the individual features of the device, and optimize resource allocation strategies. By introducing large models into ECPN, the network can efficiently integrate the idle resources of edge nodes, flexibly schedule computing tasks, improve resource utilization, meet the dynamic needs of complex tasks, and improve the overall network performance.

Enhancing Dynamic Responsiveness: In ECPN, the task requirements may change rapidly in real time, so it is crucial to enhance the dynamic response speed of the network. Traditional ML usually relies on locally limited datasets and specific scenarios for model training. When the task request changes, the model needs to be readjusted or completely retrained leading to an increase in task response time. However, the large model has excellent dynamic response capabilities in the ECPN, and can flexibly utilize the computing resources of edge nodes to adjust scheduling strategies according to real-time needs. When faced with complex task scheduling and unpredictable loads, large models exhibit efficient processing capabilities and great accuracy, which can reduce latency and ensure the stable operation of the network. This is attributed to its training on large-scale data, which empowers large models with powerful feature extraction capabilities, enabling them to quickly capture key information in dynamic environments and flexibly adapt to changing conditions and diverse input requirements.

Improving Multitask Processing Efficiency: As the variety and volume of heterogeneous tasks in the ECPN increase, the network encounters greater demands for effective multitasking. Consequently, enhancing the capability to process multiple tasks simultaneously has become crucial for ensuring the efficient operation of the network. The large model has advantages over traditional ML models in terms of task priority management and intelligent scheduling, which can reduce the waiting time between tasks and improve multitask processing efficiency. ECPN combined with a large model can efficiently manage and execute multiple tasks under limited resources, avoid resource conflicts, and reduce scheduling delays, enhancing the multitask processing capabilities and overall performance of the network.

Challenges and Solutions for Large Models in ECPN

Although incorporating large models into ECPN brings many benefits in terms of agile resource scheduling and improving task offloading performance, there are still some technical challenges in the implementation of ECPN due to insufficient service resources and increasingly stringent latency constraints.

Computation Resource Constraints: In ECPN, edge nodes are required to compute large models and process multiple heterogeneous tasks in the constraints of their limited computing resources. This dual requirement increases competition for computing resources, placing a heavier burden on the system and potentially impacting overall performance. To ensure that ECPN can process tasks and improve resource utilization, we have introduced model compression techniques such as pruning and quantization to compress large models. Quantization reduces computation and storage requirements by reducing model parameters from high to low precision. Pruning further reduces the computational complexity of the model by removing unnecessary redundant weights in the model that have less impact on the results. These two techniques enable large models to work efficiently on edge nodes with less computing resources, reducing computation burden and memory consumption in ECPN.

Real-Time Requirements: In ECPN, with the increase in the number of tasks and the dynamic changes in network conditions, it is particularly important to respond quickly and offload tasks to suitable edge nodes in real time. However, when a large model performs the complete inference process on the edge node, it faces the tradeoff between the amount of data collected and the processing speed. Collecting extensive datasets can enhance the training effectiveness of models, although it tends to slow down processing speeds. Conversely, utilizing smaller datasets accelerates model processing but may compromise the quality of training outcomes. Distributed inference provides an effective solution for this, distributing the inference task to multiple edge nodes for collaborative computing, enabling ECPN to efficiently distribute the task load and make optimal use of the computing power of multiple nodes to accelerate the processing of heterogeneous data. Each node processes a part of the tasks, effectively distributing the computational load. This distributed approach, which integrates completed inference results, alleviates the burden on individual nodes when managing heterogeneous data enhances task scheduling efficiency, and ensures real-time responses in dynamic environments.

Model Reuse and Maintenance: As task requirements and network conditions change, some edge nodes need to frequently update large models to adapt to new tasks and data in the ECPN. Incremental learning can reduces the computation resources and time costs required for model updating, while also achieving effective model maintenance, by performing local updates in steps as new environmental data is learned. Using the distributed architecture of ECPN, edge nodes can flexibly allocate computing resources based on real-time needs, and gradually adapt to new environmental data through incremental learning, further improving the efficiency of model updates, to enable direct reuse in similar scenarios without retraining. In this way, the large model-enabled ECPN can adapt to dynamic tasks and network conditions while facilitating the efficient reuse and continuous update of the model.

LLM-Enabled Privacy-Preserving Task Matching in ECPN

To demonstrate the performance of the proposed framework, this section formulates the task matching problem of ECPN in smart transportation, aiming to optimize the tradeoff between energy consumption and privacy protection.

Privacy-Preserving Task Matching Model in ECPN

In ECPN, we use graphs to describe the relationships between edge computing nodes in smart transportation, in order to capture their interactions and represent the topology of the transportation network. The graph can represent either static or dynamic connections between nodes, depending on the specific traffic scenario. Static graphs are suitable when the node layout and communication links are relatively fixed, while dynamic graphs are used when vehicles move and nodes change frequently. In the graph, nodes represent edge devices in smart transportation, edges indicate communication links or task dependencies between devices, and the state of each node includes the location, neighboring devices, and processing power. The graph captures the distribution of devices and resources in a smart transportation network, adapting to changing traffic flows, device movements, and task requirements.

For task matching in smart transportation, the tasks generated by an end device are assigned to other end devices or edge servers based on the task size, required computing resources, and latency requirements [12]. We define task matching as a one-to-many process aimed at finding the most suitable list of edge computing nodes, while applying differential privacy to task information to protect the privacy of tasks. Differential privacy [13] is a widely used privacy protection framework with a strong mathematical foundation, which protects data by introducing random noise to the query results. The noise is generated by a Laplace mechanism, and its scale is adjusted according to the sensitivity of the task information. The privacy budget controls the level of privacy protection, and the smaller the privacy budget, the higher the level of privacy protection. In our model, noise is added to the task information to protect the privacy of the task and the users involved in the matching process. During the task matching process, the strength of privacy protection is directly proportional to the amount of noise in the task information. As the degree of privacy protection increases, the smaller the privacy budget value, the more noise is introduced into the task information, which leads to an increase in errors in the matching process. The accumulation of these errors will increase the computational burden on the system, leading to an increase in energy consumption.

We formulate the task matching problem to minimize the energy consumption of task matching while protecting privacy and ensuring task requirements. Here the energy consumption of task matching is defined as the sum of

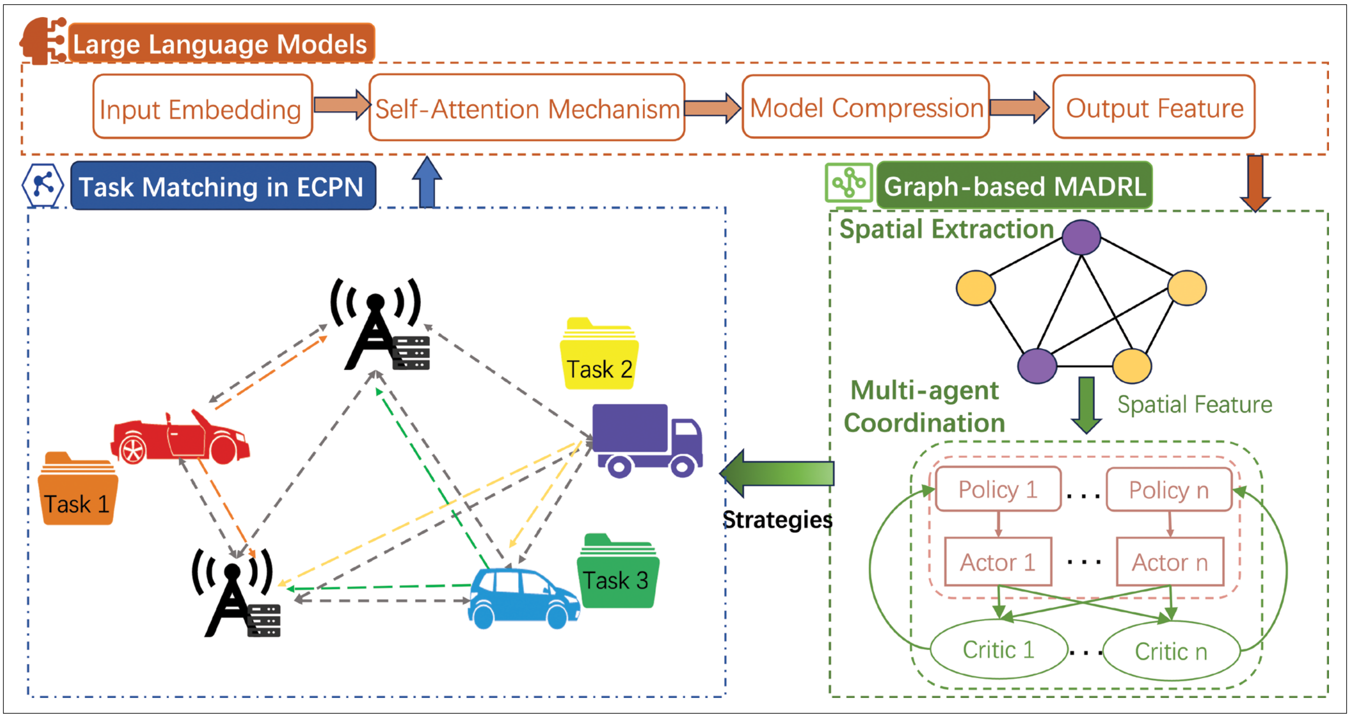

FIGURE 2. The task matching model based on LLMs and MADRL.

the energy consumption of task transmission and the energy consumption of task computation. We minimize energy consumption while protecting the privacy of tasks by adjusting chosen computing nodes, the computing resources allocated to subtasks, and the privacy budget of tasks. It is critical to note that the adjustment of the variables must comply with the following three constraints. The first is the privacy budget for tasks, which should be in the common range of differential privacy. The second one ensures that the computing resources allocated to task by edge computing node are less than the maximum value of the computing resource of node. The last constraint ensures that the total latency of task is less than the latency requirement of task.

LLMs and Graph MADRL Driven Task Matching

To address the optimization problem of task matching, we propose a model based on LLMs and graph MADRL, as shown in Fig. 2. LLMs is a type of deep learning-based artificial intelligence model that can process large-scale tasks through training on a large amount of data, which improve the efficiency of task scheduling in edge computing [14]. In task matching, LLMs provides the system with a comprehensive view of node states and task requirements by extracting global features from heterogeneous data at each edge node. These features cover the topological relationships between nodes, resource status, computing power, task priority, and the dependencies between nodes and tasks. By integrating this information, LLMs provides a basis for subsequent decision-making processes, ensuring that the network can quickly adapt and respond effectively in the face of heterogeneous tasks and dynamic environments. Although LLMs does not directly perform the task matching decision, it improves the efficiency of task matching by providing critical global information support for subsequent optimization strategies. Combining LLMs with the MADRL model, the global features extracted by LLMs provide input for the MADRL, enabling it to dynamically optimize task matching and resource allocation and identify the optimal strategy. The global feature extraction and dynamic adaptability of LLMs become the core of ECPN task matching, enhancing the flexibility and efficiency in responding to environmental changes, and improving the performance of ECPN.

The global features extracted by the LLMs are processed by a Graph Attention Network (GAT) to further capture the spatial relationships between edge nodes. Then, these features are input to the MADRL, where a network of actors and critics is used to learn the optimal task matching strategy [15]. In this process, the actor network is responsible for selecting the task matching strategy, while the critic network measures the effectiveness of the strategy by evaluating the advantages and disadvantages. With continuous exploration and adaptation, each agent explores different task allocation strategies under the changing state of edge nodes and resources. In this way, the system can dynamically adjust the matching strategy to respond to changing task requirements and resource constraints, and ensure optimal task matching and resource scheduling.

Numerical Results

In this section, we evaluate the performance of the proposed LLM-enabled energy and privacy tradeoffs task matching algorithm. We first perform global feature extraction through LLMs, which is responsible for extracting global environment information from the states of end devices and servers. These features provide key support for subsequent resource allocation and task matching. We consider simulating a real network scenario containing servers and L end devices. The location of the end is set randomly within a radius of 1000 m, and the locations are randomized. At the same time, the end devices include the numbers 5, 10, 15, and 20, respectively.

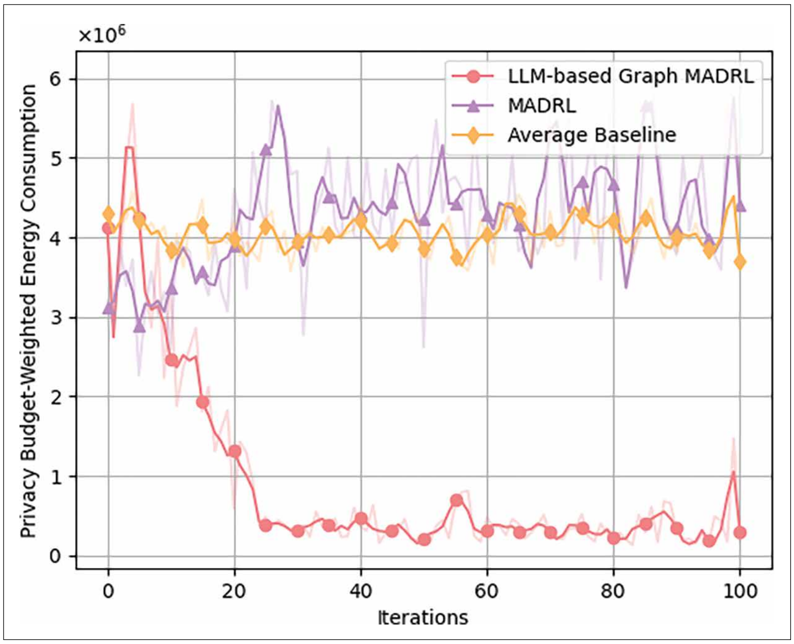

FIGURE 3. Privacy budget-weighted energy consumption in different algorithms.

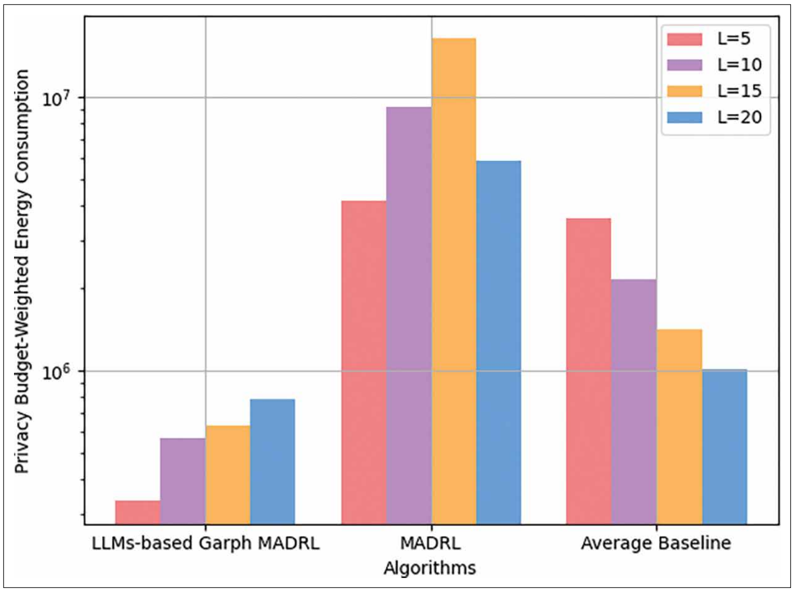

FIGURE 4. Comparison of the privacy budget weighted energy consumption for different algorithms at different numbers of devices.

The primary problem is how to balance the accuracy of the model with the constrained resources of individual edge devices as the ECPN continues to expand.

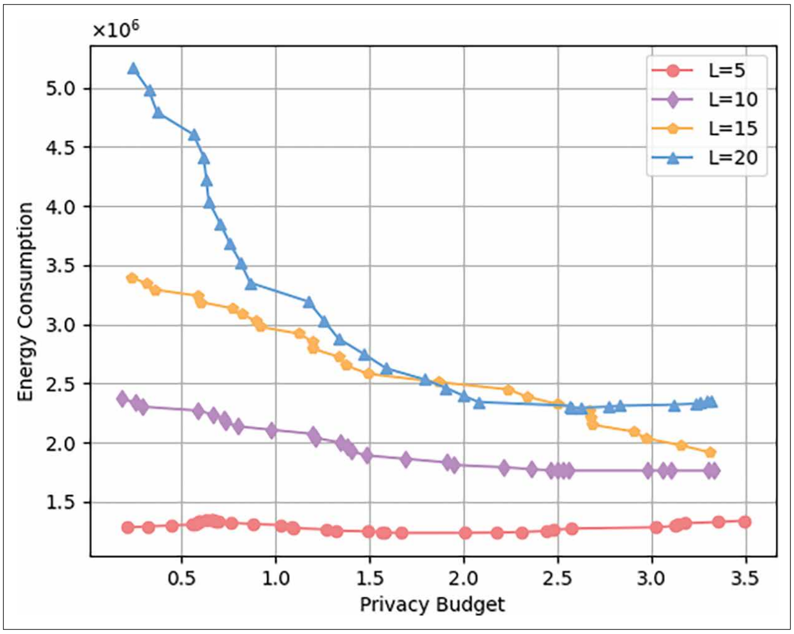

FIGURE 5. Energy consumption with privacy budget at different numbers of devices.

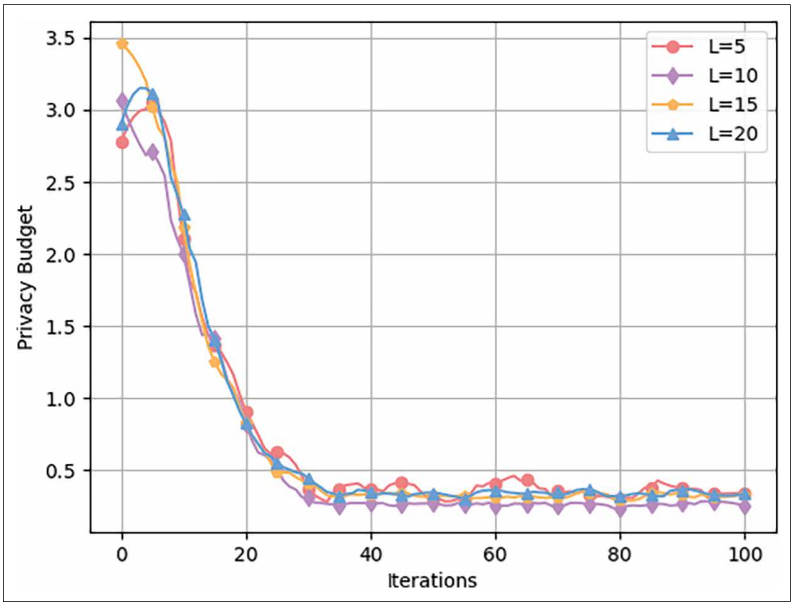

FIGURE 6. Privacy budget of tasks with different number of end device.

Fig. 3 shows the privacy budget weighted energy consumption of different algorithms including LLM-based graph MADRL, traditional MADRL and average offloading algorithms. From the figure, it can be seen that the traditional MADRL algorithm fails to converge and the energy consumption tends to increase. The reason is that the algorithm is unable to take into account the global features of the graph, which leads to slow convergence and even an increase in energy consumption. The weighted energy consumption of the average offloading algorithm fluctuates at the same height, and the energy consumption is higher than that of the proposed algorithm. In contrast, the proposed algorithm exhibits faster convergence and achieves the lowest energy consumption by effectively capturing the spatial features of the graph, which enhances task matching performance in the ECPN and reduces privacy budget-weighted energy consumption.

Fig. 4 shows the privacy budget weighted energy consumption of the three different algorithms after 100 iterations with different number of end devices. It can be seen that the proposed algorithm consistently exhibits the lowest energy consumption for all device configurations, which demonstrates its superior resource management in processing the multi-node situation. In contrast, the MADRL algorithm has not obtained an optimal task matching strategy in 100 iterations, with the highest energy consumption for all number of devices situations. The average offloading method instead performs better when the number of end devices becomes higher, but still not as good as the algorithm proposed in this paper. This result complements the previous convergence performance analysis and further confirms the benefits of the proposed algorithm in reducing energy consumption and optimizing system performance.

Fig. 5 shows the trends in the impact of privacy budget on energy consumption under different numbers of end devices. From Fig. 5, we can see that the privacy budget of the end devices is inversely proportional to the energy consumption. Energy consumption increases as the privacy budget of the end device decreases. The reason is that as the privacy budget decreases, the increase in noise introduced to protect data privacy leads to a decrease in the efficiency of computing resource allocation and an increase in computing complexity, which in turn significantly increases energy consumption. At the same time we can see that as the number of end devices increases, the energy consumption increases accordingly. The reason is that more devices require additional computing and communication resources to coordinate and manage, which leads to a higher overall system energy consumption.

Fig. 6 shows the privacy budget of tasks with different number of end device. From the figure, it can be seen that the privacy budget for tasks is relatively stable and does not fluctuate significantly with changes in the number of end devices. It shows that privacy protection strategies are based on the sensitivity of tasks, rather than being affected by the number of end devices. In addition, through this process, the system finds the most appropriate size of privacy budget, which can meet privacy protection requirements without increasing additional system overhead.

Conclusion

In this paper, we proposed a large model-enabled ECPN framework to optimize resource utilization and task processing efficiency in edge computing. The integration of large models enhanced the global feature extraction, adaptive adjustment, and multi-task parallel processing capabilities, significantly improved the capability of the system to respond to heterogeneous tasks and dynamic environments. To overcome the resource constraints of edge nodes, we used model compression, distributed inference, and incremental learning techniques to optimize the deployment of large models in ECPN. Furthermore, we propose integrating LLMs with a graph-based MADRL algorithm to develop a task matching policy that optimally balances energy consumption and privacy protection. Experimental results showed that the proposed framework and algorithm can reduce energy consumption, accelerate task matching convergence, and ensure privacy protection.

Although improvement has been made in resource allocation and large model deployment in large model-enabled ECPN, several problems remain unresolved. The primary problem is how to balance the accuracy of the model with the constrained resources of individual edge devices as the ECPN continues to expand. In the future, research should focus on improving the adaptability of large models when network conditions and task requirements continue to change. In addition, the privacy protection mechanism of distributed systems should be further developed to protect user privacy while enhancing the protection of computing power information of computing nodes.

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